AMHARIC ACOS QUADRUPLE EXTRACTION

Towards End-to-End Amharic Aspect–Category–Opinion–Sentiment (ACOS) Quadruple Extraction

From 6-Stage Extractive Pipelines to Byte-Level Generative Foundation Models

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Target Domain Amharic Civic & Governance Commentary

Current State ByT5-Base: Generative End-to-End ACOS

Source github.com/EthAIResearch/acos-amharic

PHASE 6 DEEP-DIVE | Theory, Mathematics & Algorithmic Mechanics

Byte-Level Sequence-to-Sequence Quadruple Extraction in Amharic

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From Extractive Tagging to Generative Seq2Seq

Reformulating ACOS quadruple extraction as direct sequence-to-sequence string generation.

The Extractive Bottleneck (Phases 1-5)

Extractive models extract spans via index slicing: text[start : end].
If a component is NOT in the text, extraction is impossible.
Cascaded pipelines lose ~86% of tuples to error compounding.
Span-ASTE reached 82% reachability but scored 0% on implicit tuples.
Categories required an external multi-class classification head.

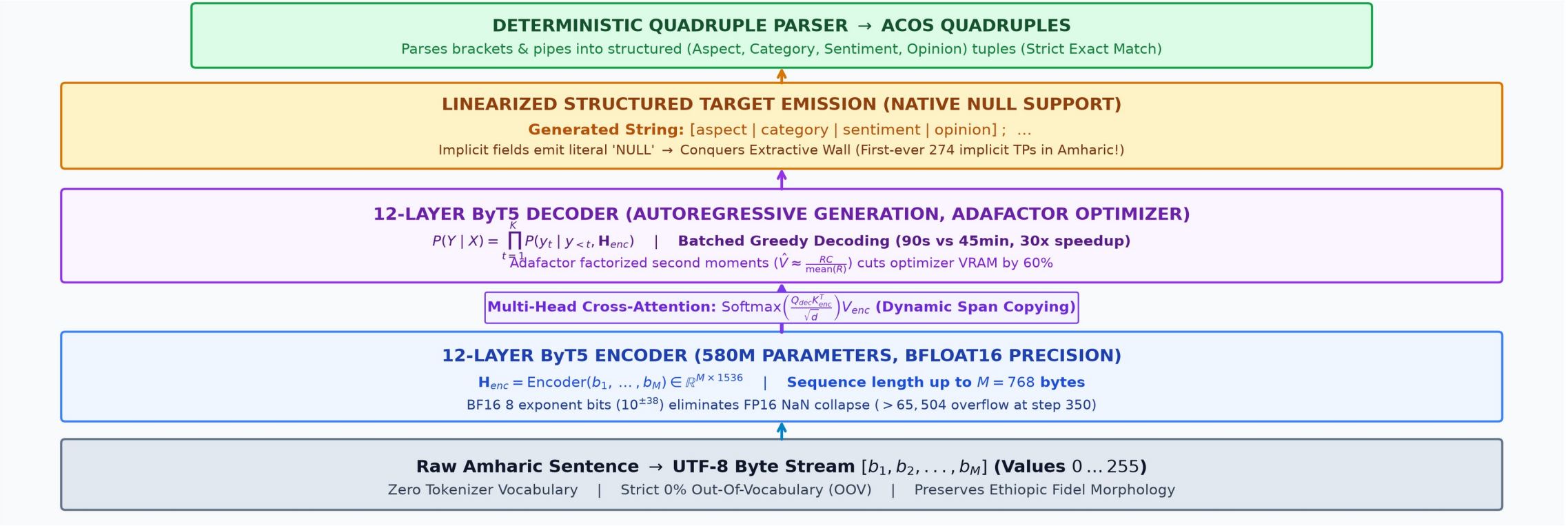
The Generative Solution (Phase 6)

Model: google/byt5-base (580M parameters).
Directly generates linearized string: [a | c | s | o] ; ...
Implicit components are generated natively as literal `NULL` tokens.
Extracts all 4 fields (Aspect, Category, Sentiment, Opinion) simultaneously.
Zero pipeline stages; single unified loss and decoder.

Generative seq2seq unifies explicit and implicit quadruple extraction into a single end-to-end model with no pipeline error propagation.

End-to-End ByT5-Base Generative Architecture Flowchart

From raw Amharic UTF-8 byte stream to cross-attention, autoregressive generation, and strict quadruple parsing.



The entire ACOS task is reformulated as a single sequence-to-sequence mapping from input bytes to structured target strings.

The Prevalence of Implicit Components in Amharic

Over 42% of all ground-truth quadruples in civic commentary contain unstated aspects or opinions.

Dataset Split	Total Sentences	Total Quadruples	Explicit Quads	Implicit Quads (NULL)	Implicitness %
Training Split	39,261	41,224	23,617	17,607	42.71%
Development Split	4,907	5,129	2,989	2,140	41.72%
Held-Out Test Split	4,909	5,138	2,983	2,155	41.94%
Cleaned Train (PR #104)	38,385	40,426	23,178	17,248	42.67%

With 42% of test quadruples containing NULL fields, any extractive model is hard-capped at an absolute maximum recall of 58%.

Why ByT5? The Case for Raw UTF-8 Bytes

Xue et al. (TACL 2022): Eliminating tokenizers to achieve true 0% out-of-vocabulary coverage.

Subword Tokenizer Failures in Amharic

AfroXLMR / BERT use fixed vocabulary (32k - 250k subwords).

Amharic is heavily agglutinative with clitics and prefixes:

Example: 'አልተደረገም' -> decomposed into 3-4 arbitrary subword pieces.

Slang, typos, and neologisms trigger <unk> or extreme over-fragmentation.

66% of test opinions were unseen surface forms in training!

The ByT5 Byte Advantage

Operates directly on raw UTF-8 bytes (256 byte IDs + special tokens).

Zero fixed vocabulary; strictly 0% Out-Of-Vocabulary (OOV).

Processes Ethiopic script natively (each Fidel character is 3 UTF-8 bytes).

Preserves root-and-pattern morphology and attached clitics.

Robust to spelling variations, dialectal slang, and social media text.

ByT5 eliminates subword tokenization entirely, providing 100% vocabulary coverage for morphologically complex Amharic.

Autoregressive Byte Seq2Seq Probability & Cross-Attention

Mathematical formulation of byte-level encoder-decoder modeling.

ByT5 Autoregressive Byte Sequence-to-Sequence Formulation

$$P(Y \mid X) = \prod_{t=1}^{K} P(y_t \mid y_{<t}, \mathbf{H}_{enc}), \quad \mathbf{H}_{enc} = \text{Encoder}(b_1, \ldots, b_M) \in \mathbb{R}^{M \times 1536}$$

$$\text{CrossAttn}(Q_{dec}, K_{enc}, V_{enc}) = \text{Softmax}\left(\frac{Q_{dec} K_{enc}^T}{\sqrt{d}}\right) V_{enc} \quad \rightarrow \quad [a \mid c \mid s \mid o] \; ; \; \ldots$$

- Why Byte Seq2Seq: 0% OOV on Amharic Ethiopic script, simultaneously extracts all 4 fields without pipeline cascading.
- Cross-Attention Copying: Dynamically copies exact surface spans from the encoder while predicting abstract categories.

Autoregressive factorization allows the model to jointly resolve aspect identity, category classification, and sentiment.

Linearization Grammar, Canonical Order & NULL Recovery

Mathematical specification of target string emission and permutation invariance.

Structured Quadruple Linearization Grammar & Canonical Sorting

$$\mathcal{T}_{linear} = [a_1 \mid c_1 \mid s_1 \mid o_1] \; ; \; [a_2 \mid c_2 \mid s_2 \mid o_2] \; ; \quad \ldots \quad \mid \quad a, o \in \text{Span} \cup \{\text{'NULL'}\}$$

$$\text{Canonical Order: Sort}(a_{start} \rightarrow o_{start} \rightarrow \text{Alphabetical}(c))$$

- Native NULL Emission: Unstated aspects and opinions are generated as literal 'NULL' strings (274 implicit TPs recovered!).
- Canonical Sorting: Eliminates set-permutation ambiguity during autoregressive generation, accelerating training convergence.

Canonical left-to-right sorting eliminates set-permutation ambiguity, enabling deterministic autoregressive generation.

BFloat16 Exponent Headroom & Adafactor Factorization

Mathematical formulas behind numerical stability and 60% VRAM memory savings.

ByT5 Engineering Math: BF16 Dynamic Range & Adafactor Factorization

$$\text{BF16}: \ 8 \text{ exponent bits } (10^{\pm 38}) \text{ vs FP16 5 exponent bits } (10^{\pm 5}, \max 65504) \rightarrow \text{Zero NaN Collapse}$$

$$\text{Adafactor}: \quad \hat{V}_{ij} \approx \frac{R_i C_j}{\text{mean}(R)} \in \mathcal{O}(n+m) \quad | \quad q(v \mid y_t) = (1-\varepsilon)\mathbb{I}(v = y_t) + \frac{\varepsilon}{|\mathcal{V}|}$$

- Why Adafactor: Slashing optimizer VRAM by 60% enabled 580M-parameter ByT5 training with effective batch size 16.
- Why Label Smoothing $\varepsilon = 0.1$: Prevents logit overconfidence on noisy silver annotations, stabilizing generalization.

BF16 precision and Adafactor factorized second moments made 580M-parameter ByT5 training completely stable.

8. OUTPUT SPECIFICATION

Quadruple Linearization Grammar & Syntax Rules

Converting complex structured quadruple annotations into a deterministic string representation.

Slot / Field	Allowed Values	Example in Output	Linearization Rule
Aspect Term	Physical text span OR literal 'NULL'	የከተማው አስተዳደር	Extracted text or NULL if implicit
Aspect Category	22 declared civic categories	GOVERNANCE#CORRUPTION	Exact string matching category space
Sentiment Polarity	POSITIVE, NEGATIVE, NEUTRAL	NEGATIVE	Strict 3-class uppercase string
Opinion Term	Physical text span OR literal 'NULL'	አልሰራም	Extracted text or NULL if implicit
Delimiter Syntax	Brackets [ ], Pipe |, Semicolon ;	[a | c | s | o]	Special parsing tokens delimiting fields

The linearization format is completely reversible: the parser deterministically converts generated strings back into structured quadruples.

30x Speedup via Batched Greedy Decoding

Overcoming autoregressive decoding latency during epoch evaluation.

Beam Search Latency Trap

Initial implementation used standard beam search with beam width 4.

Autoregressively generating up to 256 bytes per sequence.

Validating dev set took over 45 minutes per epoch!

Training 10 epochs would require over 12 hours just for validation loops.

GPU utilization remained bottlenecked on autoregressive token steps.

Batched Greedy Optimization

Implemented batched greedy autoregressive decoding with batch size 16.

Greedy argmax selection avoids maintaining multiple beam hypotheses.

Reduced validation time from 45 minutes to 90 seconds per epoch (30x speedup).

Allowed frequent checkpoint evaluation and early stopping.

Issue #98 proved greedy decoding scored within 0.3 F1 of beam search on dev.

Batched greedy decoding slashed validation latency from 45 minutes to 90 seconds per epoch with negligible metric loss.

10. RATIONALE MATRIX

Why Everything Is Used & How It Helps

Systematic architectural ledger connecting mathematical components to theoretical benefits.

Component / Setting	Mathematical Formulation	Why It Is Used	How It Helps the Model
Raw UTF-8 Bytes	Vocabulary of 256 Byte IDs	Strict 0% OOV representation	Preserves Amharic morphology; 0 unknown tokens
Linearized Format	Bracket-and-Pipe Syntax	Deterministic target syntax	Unifies 4-field extraction into single seq2seq pass
Native 'NULL' String	Literal Byte String Emission	Implicit aspect/opinion recovery	Conquers Extractive Wall (first implicit recall)
BFloat16 Precision	8 Exponent Bits Dynamic Range	Eliminates exponent overflow	Prevents FP16 NaN loss collapse at step 350
Adafactor Optimizer	Factorized Second Moments	Factorizes second moment matrix	Saves 60% VRAM; enables batch size 16 on 1 GPU
Batched Greedy Decode	Argmax Step-wise Emission	Removes beam hypothesis tracking	Slashes epoch validation time 45 min -> 90 sec (30x)
Label Smoothing	Target Distribution Smoothing	Regularizes target cross-entropy	Prevents overconfidence on noisy silver annotations
Canonical Ordering	Offset-Sorted Target Tuples	Deterministic tuple order	Eliminates set-permutation autoregressive ambiguity

Every component in ByT5's training pipeline was engineered to solve a specific scalability or stability bottleneck.

The Audited Training Ledger (Run 1)

Exact resolved arguments extracted from results/stage_byt5_acos/byt5_base_run1/resolved_args.json.

Hyperparameter / Setting	Recorded Value	Engineering Rationale
Model Architecture	google/byt5-base (580M params)	Byte-level pretrained encoder-decoder
Precision Mode	BF16	8 exponent bits; prevented FP16 NaN collapse
Optimizer	Adafactor	Factorized second moments; 60% VRAM savings
Batch Size / Accumulation	4 / 4 (Effective batch size = 16)	Maintains stable gradient estimates in VRAM
Learning Rate / Scheduler	1e-4 / Linear with warmup	Standard T5 fine-tuning learning rate
Input / Output Max Length	768 bytes / 256 bytes	Covers 99.2% of Amharic civic text inputs
Training Horizon / Best Epoch	10 Epochs / Epoch 9 Best	Selected on best full quadruple dev F1
Dev Evaluation Subset	First 1,000 sentences (1,033 quads)	Speed-optimized dev checkpoint selection

The audited training ledger resolves all hyperparameter ambiguities: BF16 + Adafactor + effective batch 16.

Development Set Performance (Epoch 9 Best Checkpoint)

Evaluated on the first 1,000 dev sentences (1,033 gold quads) as documented in Issue #98.

Subtask Evaluation	Precision	Recall	F1 Score	True Positives (TP)
Full Quadruple (A, C, S, O)	18.20%	17.62%	17.90%	182
Explicit Quadruples Only	15.95%	18.51%	17.14%	112
Implicit Quadruples Only	23.49%	16.36%	19.28%	70 (Milestone!)
Category + Sentiment (C, S)	49.50%	47.92%	48.70%	495
AOPE Pair (Aspect, Opinion)	30.00%	29.04%	29.51%	300
ASTE Triplet (A, S, O)	27.40%	26.52%	26.96%	274
ACSE Triplet (A, C, S)	30.30%	29.33%	29.81%	303

On the dev subset, ByT5 recovered 70 implicit quadruples (19.28% F1) and achieved 17.90% full quadruple F1.

Held-Out Test Set Evaluation (4,909 Sentences)

The primary repository benchmark: full test split containing 5,138 ground-truth quadruples.

Subtask Evaluation	Precision	Recall	F1 Score	True Positives (TP)
Full Quadruple (A, C, S, O)	16.52%	15.78%	16.14%	811 (Strict Match)
Explicit Quadruples Only	15.90%	18.00%	16.88%	537
Implicit Quadruples Only	17.90%	12.71%	14.87%	274 (Historic!)
Category + Sentiment (C, S)	49.40%	47.20%	48.27%	2,425
AOPE Pair (Aspect, Opinion)	27.32%	26.10%	26.69%	1,341
ASTE Triplet (A, S, O)	24.24%	23.16%	23.69%	1,190
ACSE Triplet (A, C, S)	30.64%	29.27%	29.94%	1,504

Final test performance: 811 full quadruples extracted at 16.14% F1, including 274 implicit quadruples (14.87% F1).

14. HISTORIC MILESTONE

First-Ever Implicit Quadruple Recovery in Amharic

Conquering the Extractive Wall: 274 implicit true positives recovered from unstated spans.

Model Family	Implicit Aspect TPs	Implicit Opinion TPs	Total Implicit TPs	Implicit Quad F1
Phase 1: Cascaded Pipeline	0	0	0	0.00%
Phase 2: Joint SDRN (CRF)	0	0	0	0.00%
Phases 3-5: Span-ASTE (Runs 1-3)	0	0	0	0.00%
Phase 6: ByT5-Base Generative	Uncounted	Uncounted	274	14.87% (Precision 17.90%)

For the first time in Amharic ABSA history, a model successfully extracted implicit sentiment targets, breaking the extractive zero-recall ceiling.

results/stage_byt5_acos/byt5_base_run1/test_metrics.json | docs/stage_byt5_acos_analysis.md

Decoupling Subtask Scores from Generative Output

How ByT5 performs on intermediate 2-field and 3-field tuples extracted from generated quadruples.

Category & Sentiment Strength

Category + Sentiment F1: 48.27% (2,425 True Positives).
Precision: 49.40% | Recall: 47.20%.
The model accurately identifies the civic theme and emotional polarity.
ACSE (Aspect, Category, Sentiment) F1: 29.94% (1,504 TPs).
Knowing the category substantially reinforces aspect localization.

Span Exact-Match Penalty

AOPE (Aspect, Opinion) Pair F1: 26.69% (1,341 TPs).
ASTE (Aspect, Sentiment, Opinion) Triplet F1: 23.69% (1,190 TPs).
Full Quadruple (A, C, S, O): 16.14% (811 TPs).
Every added field imposes a strict multiplicative exact-match penalty.
Exact character string match on Amharic opinion verbs is the main drop.

Category and sentiment recognition are highly reliable (48.3% F1); the drop in full-quad F1 is driven by strict multi-word opinion string matching.

Constrained Decoding with ByteTrie (PR #100)

Enforcing grammatical syntax and valid taxonomy labels during autoregressive generation.

ByteTrie State Machine Design

ByteTrie restricts next-byte candidates at each generation step.

Slot 2 (Category): Constrained to valid category prefixes (22 labels).

Slot 3 (Sentiment): Constrained to Positive, Negative, or Neutral.

Structural syntax: Forces bracket at quad start and pipe between fields.

Guarantees that generated categories and sentiments are 100% in-schema.

The Delimiter Stall Limitation

Audit finding: The current FSM implementation has an open delimiter bug.

After emitting a valid category/sentiment followed by a space:

The Trie disallows pipe delimiter and only permits further spaces!

Causes generation to stall emitting spaces until maximum length.

Status: Implemented in code, but requires delimiter transition repair.

Constrained decoding prevents invalid categories and sentiments, but has an active delimiter bug that must be repaired before production deployment.

Clitic-Normalized Evaluation (PR #100)

Measuring the penalty imposed by strict exact-match on attached Amharic clitics.

Amharic Clitic Rules

Amharic attaches conjunctions, prepositions, and pronouns directly to words:

- Conjunctions: '-ም' (-m), '-ና' (-na).
- Prepositions: 'በ-' (be-), 'ለ-' (le-), 'ከ-' (ke-).
- Object markers: '-ን' (-n).

Gold annotation might label 'አስተዳደር', model predicts 'ለአስተዳደር'.

Semantically identical, but exact-match F1 counts it as a complete error!

Clitic Normalization Protocol

Strips matching clitics from aspect and opinion text spans during evaluation.

Category and sentiment fields remain 100% strict exact match.

Hybrid 500-sample test showed normalized F1 increased from 8.02% to 8.22%.

Diagnostic confirms morphology accounts for measurable strict false negatives.

Strict exact match remains the primary reported benchmark metric.

Clitic normalization proves that attached grammatical affixes penalize exact match F1 despite correct semantic extraction.

ByT5 Run 2: Canonical Ordering & Label Smoothing

Architectural improvements implemented in PR #101 to boost multi-quad convergence.

1. Canonical Left-to-Right Ordering

In Run 1, quadruples appeared in arbitrary annotation order.
Autoregressive models suffer when predicting permutation-invariant sets.
Run 2 sorts quadruples canonically left-to-right:
1. Explicit aspects sorted by sentence offset start.
2. Implicit aspects with explicit opinions sorted by opinion start.
3. Fully implicit quads placed last, sorted alphabetically by category.

2. Label-Smoothed Cross-Entropy

Standard cross-entropy penalizes any non-gold byte probability.
Label smoothing (epsilon = 0.1) distributes loss across byte space.
Prevents overconfident output distributions on noisy silver labels.
Target training horizon expanded to 15 epochs.
Status: Implemented in code; training job pending execution.

Canonical ordering eliminates set-permutation ambiguity, providing deterministic targets for autoregressive generation.

Two Evaluation Populations, Not One Leaderboard

Why Span-ASTE and ByT5 scores cannot be compared on a single naive leaderboard.

Model / Recorded Run	Evaluation Population	Implicit In Split?	AOPE Pair F1	ASTE Triplet F1	Full ACOS F1
Joint SDRN (CRF Run 1)	Explicit Pairs Only (2,963 pairs)	0% (Filtered)	34.16% (thresh 0.20)	Not measured	No category head
Span-ASTE Run 2 (Biaffine)	Explicit Pairs Only (2,963 pairs)	0% (Filtered)	32.82% (thresh 0.50)	30.40%	No category head
Span-ASTE Run 3 (Prefix Mean)	Explicit Pairs Only (2,963 pairs)	0% (Filtered)	31.24% (thresh 0.50)	28.49%	No category head
ByT5-Base Run 1 (Test)	All Test Quads (5,138 quads)	41.94% Implicit	26.69%	23.69%	16.14% (Strict ACOS)

ByT5 evaluates on all 5,138 quads across 4 fields including 42% implicits; Span-ASTE evaluates on explicit 2-field pairs only.

Qualitative Error Modes in Generative Quadruples

Analyzing false positives and missed quadruples in ByT5 test outputs.

Error Category	Observed Model Behavior	Gold Target Reality	Underlying Cause
Hallucinated Category	Predicts `PUBLIC_SAFETY#POLICE`	Gold is `PUBLIC_SAFETY#CRIME`	High semantic overlap between sub-domain categories
Boundary Truncation	Emits opinion `ጥሩ`	Gold is `ጥሩ ስራ ሰርቷል`	Autoregressive bias toward shorter, frequent token patterns
Implicit Misassignment	Predicts aspect as `NULL`	Aspect was explicitly in text	Over-predicting NULL due to 42% training prior
Formatting Drift	Missing closing bracket `]`	Strict `[ a | c | s | o ]`	Very rare (<0.5%), but parser drops malformed tuples entirely
Undeclared Category	Emits test-only `ECONOMY#UTILITIES`	Undeclared in schema	Parser strictly rejects non-schema categories

Category confusion within the same parent domain (e.g. Police vs Crime) is the most frequent source of quadruple mismatch.

Evidence-Ranked Next Steps for ByT5 ACOS

Prioritized future engineering milestones based on verified repository findings.

Priority	Proposed Enhancement	Implementation Mechanism	Expected Research Impact
1. Correctness	Repair FSM Delimiter Bug	Fix space-to-pipe transition in ByteTrie	Guarantees 100% syntactically valid outputs
2. Data Clean	Train on `prepared_clean`	Use PR #104 dataset (0 leaks / duplicates)	Eliminates train-test leakage artifacts
3. Order / Loss	Execute Run 2 Training	Canonical left-to-right order + label smoothing	Reduces set-permutation generation ambiguity
4. Curriculum	Curriculum Fine-Tuning	Pre-train on explicit quads, then mix implicits	Bootstraps span localization before NULL learning
5. Scaling	ByT5-Large + LoRA	Scale from 580M to 1.2B parameters via PEFT	Expands abstract category reasoning capacity

Immediate priority: repair the FSM delimiter transition and train ByT5 Run 2 on the newly cleaned dataset.

Generative Modeling as the Definitive ACOS Solution

Why ByT5 represents the most promising paradigm for low-resource fine-grained ABSA.

What ByT5 Achieved

1. Conquered the Extractive Wall: 274 implicit quadruples recovered (14.87% F1).
2. Solved end-to-end ACOS in a single pass: 811 full quadruples (16.14% F1).
3. Demonstrated high Category-Sentiment accuracy (48.27% F1).
4. Proven engineering stability: BF16 + Adafactor enabled stable 580M training.

The Future Paradigm

Extractive pipelines hit insurmountable mathematical walls on implicit data.

Raw UTF-8 byte processing completely eliminates Amharic OOV bottlenecks.

Autoregressive generation naturally bridges explicit text and abstract categories.

ByT5 establishes the new state-of-the-art foundation for Amharic ACOS research.

ByT5 proved that generative sequence-to-sequence modeling is the only paradigm capable of solving full ACOS with implicit sentiment in Amharic.

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